

# **PROJECT PLAN REPORT**

## **WBS, Gantt Chart, Milestones and Scheduling**

Development of a Learning Management System (LMS) for Online Education

Muhammad Huzaifa Kamran

66772

### **1. Introduction**

This report showcases the work assigned to me during the LMS Project. The focus is on:

- breaking the project into manageable work packages
- arranging them into a realistic schedule
- identifying major milestones
- showing the dependencies between activities through a Gantt chart and activity sequence table.

The LMS project aims to develop a browser-based online education system that supports students, teachers and administrators.

The planned modules include:

- user login
- course management
- student and teacher dashboards
- assignment and quiz management
- attendance tracking
- digital gradebook
- notice board
- discussion forums

### **2. Scheduling (This is tentative to changes based on unforeseen circumstances)**

- The project schedule is planned as a one-month timeline starting from 1 May 2026 till 31 May 2026.
- Development activities follow a sequential-plus-overlap approach, where design and database work can run partly in parallel but testing starts only after integrated modules are ready.

### 3. Work Breakdown Structure (WBS)

The representation below uses three levels: major project phase, work package and key output.

| <b>WBS ID</b> | <b>Work Package</b>         | <b>Description</b>                                                             | <b>Main Output</b>                            |
|---------------|-----------------------------|--------------------------------------------------------------------------------|-----------------------------------------------|
| 1.0           | Project Initiation          | Confirm project aim, high-level requirements and approval to proceed.          | Project charter and initial requirement notes |
| 2.0           | Planning and Scope Baseline | Define scope, deliverables, exclusions, WBS and schedule baseline.             | Approved WBS and baseline timeline            |
| 2.1           | Requirements Review         | Review student, teacher and administrator needs for the LMS.                   | Prioritised requirement list                  |
| 2.2           | WBS Development             | Break LMS scope into work packages and assign task IDs.                        | WBS table and activity list                   |
| 2.3           | Timeline Planning           | Estimate activity duration, dependencies and project milestones.               | Gantt chart and milestone schedule            |
| 3.0           | System Design               | Prepare UI/UX flow, system architecture and database structure.                | Design specifications                         |
| 3.1           | UI/UX Design                | Plan screens for login, student dashboard, teacher dashboard and admin panel.  | Wireframes and UI flow                        |
| 3.2           | Database Design             | Design tables for users, courses, assignments, quizzes, attendance and grades. | Database schema                               |
| 4.0           | Development                 | Build the core LMS web application and modules.                                | Working LMS modules                           |
| 4.1           | Authentication Module       | Implement login, logout, roles and access control.                             | Secure access system                          |
| 4.2           | Course Management           | Implement course creation, enrolment and resource upload.                      | Course management module                      |
| 4.3           | Dashboards                  | Build student, teacher and administrator dashboards.                           | Dashboard interfaces                          |
| 4.4           | Assessment Modules          | Build assignments, quizzes, attendance and gradebook workflows.                | Assessment and tracking modules               |
|               | Communication               | Build notice board,                                                            | Communication                                 |

|     |                         |                                                                      |                               |
|-----|-------------------------|----------------------------------------------------------------------|-------------------------------|
| 4.5 | Modules                 | discussion forum and notifications.                                  | features                      |
| 5.0 | Testing                 | Test modules individually and as an integrated system.               | Test report and fixed defects |
| 5.1 | Integration Testing     | Check whether modules work together correctly.                       | Integration test results      |
| 5.2 | User Acceptance Testing | Validate the LMS with expected student, teacher and admin workflows. | UAT approval                  |
| 6.0 | Deployment and Closure  | Deploy the final LMS, prepare documentation and close the project.   | Live LMS and closure record   |

#### 4. Activity List, Duration and Dependencies

Dependencies show which activities must be completed before starting another activity can begin. This is important because modules such as dashboards, assignments and quizzes cannot be completed until authentication, database and course management foundations are ready.

| ID | Activity                                       | Duration | Start       | Finish      | Predecessor(s) | Milestone |
|----|------------------------------------------------|----------|-------------|-------------|----------------|-----------|
| A1 | Project initiation and charter confirmation    | 2 days   | 01 May 2026 | 02 May 2026 | -              | No        |
| A2 | Requirements collection and stakeholder review | 3 days   | 03 May 2026 | 05 May 2026 | A1             | No        |
| A3 | Scope baseline and feature prioritisation      | 1 day    | 06 May 2026 | 06 May 2026 | A2             | Yes       |
| A4 | WBS finalisation and work-package approval     | 1 day    | 07 May 2026 | 07 May 2026 | A3             | Yes       |
| A5 | UI/UX design for LMS screens                   | 4 days   | 08 May 2026 | 11 May 2026 | A4             | No        |
| A6 | Database design and data structure planning    | 4 days   | 08 May 2026 | 11 May 2026 | A4             | No        |

|     |                                                        |        |             |             |            |     |
|-----|--------------------------------------------------------|--------|-------------|-------------|------------|-----|
| A7  | GitHub/Jira project setup for scheduled work tracking  | 2 days | 08 May 2026 | 09 May 2026 | A4         | No  |
| A8  | Authentication and role-based access module            | 4 days | 12 May 2026 | 15 May 2026 | A5, A6, A7 | No  |
| A9  | Course management module                               | 4 days | 16 May 2026 | 19 May 2026 | A8         | No  |
| A10 | Student, teacher and admin dashboards                  | 4 days | 20 May 2026 | 23 May 2026 | A9         | No  |
| A11 | Assignments, quizzes, attendance and gradebook modules | 4 days | 24 May 2026 | 27 May 2026 | A10        | No  |
| A12 | Notice board, notifications and discussion forum       | 3 days | 25 May 2026 | 27 May 2026 | A10        | No  |
| A13 | Module integration and internal review                 | 2 days | 28 May 2026 | 29 May 2026 | A11, A12   | Yes |
| A14 | System testing and user acceptance testing             | 2 days | 30 May 2026 | 31 May 2026 | A13        | Yes |
| A15 | Final deployment and project closure                   | 1 day  | 31 May 2026 | 31 May 2026 | A14        |     |

## 5. Milestone Schedule

Milestones are zero-or-near-zero decision points used to check progress. They do not represent full work packages themselves; they mark important approvals or completion points in the schedule.

| Milestone ID | Milestone                          | Target Date | Completion Criteria                                                                                       |
|--------------|------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------|
| M1           | Scope baseline approved            | 06 May 2026 | Requirements and project boundaries are agreed.                                                           |
| M2           | WBS and schedule baseline approved | 07 May 2026 | Work packages, activity list and May timeline are confirmed.                                              |
| M3           | Core LMS modules completed         | 27 May 2026 | Authentication, course management, dashboards, assignments, quizzes, attendance and communication modules |

are completed.

|    |                                           |             |                                                                                                   |
|----|-------------------------------------------|-------------|---------------------------------------------------------------------------------------------------|
| M4 | Integration and internal review completed | 29 May 2026 | Major LMS modules are connected and internally reviewed.                                          |
| M5 | Testing, deployment and closure completed | 31 May 2026 | Testing is completed, the LMS is ready for final submission/deployment and the project is closed. |

## 6. Gantt Chart

Overlapping tasks are used where possible to shorten the project duration, but major dependent activities still follow a controlled sequence.

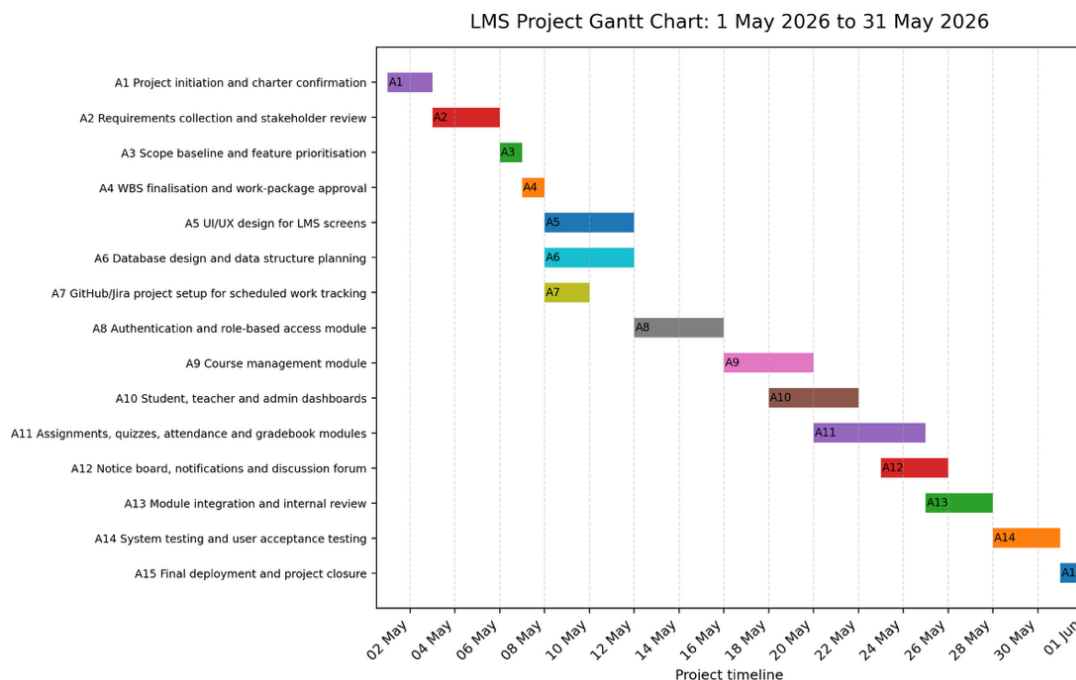


Figure 1: WBS-based Gantt chart for the LMS project

## 7. Interdependencies Between Activities

| Dependency    | Reason                                                                              |
|---------------|-------------------------------------------------------------------------------------|
| A1 → A2       | Requirements collection starts after project initiation is confirmed.               |
| A2 → A3       | The scope baseline can only be approved after requirements are reviewed.            |
| A3 → A4       | The WBS must be finalised after the scope boundary is clear.                        |
| A4 → A5/A6/A7 | UI design, database planning and GitHub/Jira setup start after the WBS is approved. |

|                      |                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------|
| A5 + A6 + A7<br>→ A8 | Authentication development depends on UI flow, database structure and environment setup.                 |
| A8 → A9              | Course management needs role-based access and user authentication first.                                 |
| A9 → A10             | Dashboards depend on the course management module because dashboards display course-related data.        |
| A10 → A11/A12        | Assignments, quizzes, attendance and communication modules depend on dashboard workflows and user roles. |
| A11 + A12 →<br>A13   | Module integration starts after academic and communication modules are completed.                        |
| A13 → A14            | System testing and UAT are performed after the integrated review is complete.                            |
| A14 → A15            | Final deployment and closure occur after testing and user acceptance are completed.                      |

## 8. Schedule Monitoring Approach

| Monitoring Action      | Purpose                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Weekly progress review | Check completed, in-progress and delayed tasks against the activity table.                                                          |
| Milestone tracking     | Confirm whether each milestone has been achieved by its target date.                                                                |
| Dependency control     | Identify tasks blocked by unfinished predecessor activities.                                                                        |
| Schedule update        | Update the Gantt chart when approved changes affect activity dates, durations or the fixed project deadline of <b>31 May 2026</b> . |
| Escalation             | Report delays that threaten testing, UAT or deployment dates.                                                                       |

## 9. Conclusion

The project report explain the scheduling plan for the LMS project:

- The WBS Divides the work into separate work packages

- the activity table showcases the project duration
- dependencies of multiple processes
- the milestone schedule clarifies major control points
- the Gantt chart gives a clear visual timeline.

This planning structure will help the project team coordinate development work, monitor progress and complete the LMS within the planned 1 month period.

## Appendix A: Simplified WBS Hierarchy

### 1.0 Project Initiation

### 2.0 Planning and Scope Baseline

#### 2.1 Requirements Review

#### 2.2 WBS Development

#### 2.3 Timeline Planning

### 3.0 System Design

#### 3.1 UI/UX Design

#### 3.2 Database Design

### 4.0 Development

#### 4.1 Authentication Module

#### 4.2 Course Management

#### 4.3 Dashboards

#### 4.4 Assessment Modules

#### 4.5 Communication Modules

### 5.0 Testing

#### 5.1 Integration Testing

#### 5.2 User Acceptance Testing

### 6.0 Deployment and Closure